

SONAL GUPTA

Automation Test Engineer

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Professional Summary:

- **Work Authorization: Holding a valid USA L2 Visa.**
- 14+ years of experience in the industry, including 10 years at Siemens India.
- Diligent Test Automation Lead with solid foundation in leading automation testing efforts.
- Hands-on experience in Functional testing, System testing, Integration testing, Performance testing, Regression testing and User Acceptance testing.
- Successfully managed end-to-end automation processes, enhancing test coverage and reducing manual effort.
- Demonstrated expertise in scripting and framework development, manual testing as well as strong analytical and troubleshooting skills.
- Sub-system Architect for the TGMT project and Lead Test Automation Engineer for a Scrum team.

Technical Skills:

- Milestone planning
- Release management
- Test strategy development
- C/C++, Linux
- HP ALM, HP UFT
- GIT, CI/CD, JIRA, RTC, DOORS

Professional Experience:

Siemens Technology And Services, Pune, India | May 2014 – Nov 2024

Project #1 (Automation Test Lead):

Responsibilities:

- Worked as test automation lead for a team of 6 international Engineers to test a safety software for the onboard automatic train protection component (CBTC, GoA4, SIL4, EN50128).
 - Acted as scrum master and liaison between Pune and France teams.
 - Developed test plans and test strategies, performed test optimization, and created test specifications.
 - Documented test scenarios and test steps using tools like JIRA, HP ALM, or DOORS, ensuring traceability to functional requirements.
 - Converted manual test cases into automated scripts using HP UFT, leveraging its VBScript-based environment to streamline repetitive testing tasks and improve test coverage.
 - Developed and executed automation test scripts using HP UFT (Unified Functional Testing), ensuring comprehensive validation of application functionality and regression scenarios.
 - Utilized HP ALM for test management, defect tracking, and reporting to maintain streamlined communication across project stakeholders.
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- Documented and tracked defects systematically in HP ALM, providing detailed descriptions, steps to reproduce, and severity classifications.
- Created reports and graphs based on logged defects in HP ALM to communicate the defect status to stakeholders.
- Worked on User Acceptance testing by designing and executing test cases based on user requirements, and ensuring the product meets business needs from an end-user perspective.
- Developed and implemented test plans for failover scenarios, verifying system resilience and high availability during unexpected failures.
- Conducted troubleshooting and root cause analysis, collaborating with development teams to resolve critical issues identified during various testing phases.
- Designed and executed automated test scripts using tools like HP UFT, streamlining regression and smoke testing processes.
- Performed root cause analysis for recurring incidents, implementing long-term fixes to prevent future occurrences.
- Worked closely with business analysts and SMEs to gather and understand business requirements, ensuring comprehensive test coverage aligned with project goals.
- Created and executed functional test cases to validate core business functionalities, ensuring alignment with requirements and user expectations.
- Developed and managed regression test suites to ensure seamless functionality after code changes and software updates.
- Performed regression testing on resolved defects to validate fixes and ensure no adverse effects on existing functionality.
- Facilitated joint application design (JAD) sessions with cross-functional teams to define acceptance criteria and ensure alignment between stakeholders.
- Automated regression test scripts using tools like HP UFT, reducing manual testing efforts and enhancing test coverage.
- Validated test scripts against system requirements by conducting requirement traceability analysis to ensure comprehensive coverage.
- Collaborated with technical staff to identify system dependencies and constraints, translating technical requirements into actionable test plans.
- Developed and executed performance test scripts using LoadRunner to evaluate application scalability, reliability, and responsiveness under varying load conditions.
- Participation in design decision during client call.
- Developed detailed test plans and strategies based on functional specifications and user stories, covering end-to-end workflows for web applications.
- Designed and maintained reusable automation frameworks using HP UFT, enabling modular and data-driven testing approaches for scalable test execution.
- Implemented object repository models (shared and local) in HP UFT to improve maintainability and reduce redundancy in test scripts.
- Implemented test automation using C++, conducted commissioning, and managed RTC and DOORs.
- Specialized in OBCU sub-system testing and HMI testing.
- Provided application support by monitoring, identifying, and resolving issues in production and non-production environments.
- Developed and executed complex SQL queries to validate backend data and ensure database integrity during functional and regression testing.
- Performed performance tuning and optimization of SQL queries for improved test efficiency and reduced execution time.
- Created detailed test cases to validate database constraints, relationships, and transactions in relational database systems.
- Worked on CFX analysis, test plans, and feature development as a sub-system architect.

Project #2 (Senior Software Engineer):

Responsibilities:

- Worked on a scratch project which provide connectivity between BACnet devices and cloud Connectivity.
- Participated in user requirement understanding from UML sequence diagram and participated in design model.
- Implemented features in C/C++ and linux and performed debugging using GDB.
- Performed unit testing, system testing, Valgrind testing.
- Performed keyword-driven and data-driven testing by building reusable components in HP UFT to ensure efficient test execution for varying input datasets.
- Automated database testing by writing PL/SQL scripts for data validation, stored procedures, and triggers.
- Built modular and data-driven test scripts, enabling easy reusability across multiple test cases and applications.
- Developed custom libraries and utilities to handle repetitive tasks, such as data input, validation, and logging, improving script robustness.
- Maintained a repository of reusable functions, ensuring consistency and efficiency across projects.
- Collaborated with QA and development teams to identify automation opportunities and create automation plans based on functional test cases.
- Documented detailed issue resolution steps to build a knowledge base, enabling quick turnaround for similar future issues.
- Responded to on-call support requests, addressing critical incidents and minimizing downtime during off-hours.
- Created and maintained detailed technical documentation, including test plans, test cases, defect reports, and troubleshooting guides.
- Collaborated with developers and business analysts to prioritize and resolve defects, ensuring timely delivery of quality applications.
- Updated test scripts in HP ALM to reflect changes in business requirements or application functionality.
- Prepared and shared defect status reports with stakeholders, highlighting critical issues and tracking progress toward resolution.
- Facilitated defect triage meetings, ensuring alignment on defect prioritization and resolution timelines.
- Enhanced testing efficiency by identifying and documenting edge cases missed in initial test scripts, improving overall application quality.
- Improved cross-team communication through clear and consistent defect reporting and follow-ups within HP ALM.
- Monitored application performance using LoadRunner's integrated monitoring tools, capturing key metrics such as response time, throughput, and resource utilization.
- Participation in design decision during client call.
- Performed user documentation creation related to feature provided for a release.

Senior Software Engineer | GlobalLogic, Noida, India | Feb 2010 – Nov 2013

Responsibilities:

- Worked on session Border Controller (SBC) project that manages and secures Voice over Internet Protocol (VoIP) communication.
 - Participated in user requirement understanding from user requirement manual.
 - Validated automated scripts by performing dry runs and ensuring alignment with business requirements and acceptance criteria.
 - Created detailed automation reports and dashboards to provide stakeholders with insights into test coverage, defect trends, and execution results.
 - Designed and implemented performance test scenarios to evaluate application scalability, stability, and responsiveness under peak load and stress conditions.
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- Designed parameterized test scripts to handle dynamic data inputs and simulate realistic user interactions with applications.
- Created robust and reusable test cases to validate functionality, usability, and performance of web applications, ensuring adherence to business requirements.
- Managed test artifacts such as test plans, test cases, and scripts, ensuring version control and traceability throughout the project lifecycle.
- Collaborated with cross-functional teams, including developers and business analysts, to define test objectives and resolve issues efficiently.
- Documented application workflows, configurations, and testing procedures, ensuring clarity and consistency for team members and stakeholders.
- Designed and implemented test cases in C/C++ to validate algorithms and system modules in a software development lifecycle.
- Developed unit tests using C++ frameworks like Google Test to verify functionality and ensure code coverage.
- Analyzed memory management and performance issues in C/C++ applications to identify and resolve defects.
- Integrated automation scripts with manual testing strategies for end-to-end validation of software products.
- Participated in code reviews and contributed to improving the quality and maintainability of PL/SQL, SQL, and C/C++ codebases.
- Improved system stability by conducting rigorous failover and performance testing, identifying critical vulnerabilities before production deployment.
- Designed test cases for positive, negative, edge case, and boundary condition scenarios to ensure comprehensive application validation.
- Participated in design model and implemented features using C/C++ and Linux.
- Performed unit testing and worked on regression bugs on different releases. Worked on Coverity tool.
- Migrated codebase from CVS to GIT version control as a single contributor.

Educational Details:

- Bachelor of Engineering in Computer Science from ITM Universe (RGPV Bhopal), Gwalior India.
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